Retrieval that Cannot See the Future: Temporal Evaluation of Fact-Checked Claim Retrieval Project Proposal for NLP Course, Winter 2026/27 (revised after Phase 1 feedback; point-by-point answers in `rebuttal.pdf`)

Paweł Pozorski **Mateusz Jarosz**

Paweł Florek **Jan Cwalina**

supervisor: Kuntur Soveatin
Warsaw University of Technology

Template: Anna Wróblewska (Warsaw University of Technology), adapted from the ACL 2015 conference formatting template.

Abstract

Previously fact-checked claim retrieval (PFCR) ranks an archive of fact-checks for a social media post. Published benchmarks search the full archive, including fact-checks written after the post, while a fact-checker at post time can only search what already exists. We propose a *causal* evaluation protocol: for each post q with time t_q, the pool is $P(q) = \{f : t_f < t_q\}$, and we report Success@10 (fraction of queries with a gold fact-check in the top 10) and mean reciprocal rank (MRR) under random, size-matched random, and strict-causal pools on MultiClaim, with Polish as focus and English as control. Deliverables: the protocol and rebuildable split as id lists and code; a results table of at least four systems under both protocols; and a gap decomposition with 50 hand-coded newly failing queries.

1 Scientific goal

Problem. PFCR matches a post q to earlier fact-checks f that already debunk the same claim (Shaar et al., 2020; Pikuliak et al., 2023). Standard evaluation lets q retrieve fact-checks published after t_q, which rewards matching a later article that quotes the post. We ask how much reported performance survives a time-respecting index, and why it drops.

Research questions. RQ1: How much Success@10 (S@10) and mean reciprocal rank (MRR) survive a causal index (unconditionally and on answerable queries), and does system ranking change? **RQ2:** How much of the random-vs-causal gap is answerability loss, pool-size effects, training leakage, or missing labels for earlier sibling debunks, and how does the remaining gap relate to the time lag between post and debunk, lexical overlap and narrative drift (covariates)? **RQ3:** Is the causal penalty larger for Polish than for English, and how much do earlier English fact-checks recover for Polish posts?

Hypotheses (prespecified, not formally preregistered). Most MultiClaim gold pairs are post-before-fact-check (H1); on answerable queries, time-local distractors are harder than a size-matched random pool (H2); Polish shows a larger causal penalty than English, especially for lexical BM25 (H3); fine-tuned retrievers lose more under chronological training than zero-shot ones (Shaar et al., 2022) (H4).

Whether large language model (LLM) rerankers gain on facts seen in pretraining is left to future work. Language and retrieval bias across languages is analysed by Vykopal et al. (2026) and is out of scope here; we only compare each language with itself over time. Random splits overestimate performance (Søgaard et al., 2021), which motivates the time-respecting split.

2 Significance and state of the art

We summarise fifteen prior works. Table 1 lists the published numbers we compare on one 0–1 scale, with the dataset and the split or evaluation condition named for every value ("no split" where the source has none). No PFCR evaluation we found restricts the fact-check index *per query* to docu-

ments published before the post.

PFCR task and shared tasks. Shaar et al. (2020) define PFCR as ranking verified claims for a social-media or debate input; on Snopes tweets and PolitiFact debate sentences, BM25 plus a ranking support vector machine (rankSVM) reaches MRR 0.788 and S@10 0.915 on the Snopes test (BM25 top-50 rerank). Shaar et al. (2021) run Task 2 of CheckThat! 2021 at the Conference and Labs of the Evaluation Forum (CLEF) (English/Arabic tweets; English debates) with fixed pools and no date filter; best English tweet MAP@5 is 0.883 (Aschern) vs BM25 0.749 on the 70/15/15 tweet split. Nakov et al. (2022) rerun the lab in 2022 with a small new test set; RIET Lab reaches MAP@5 0.956 on English tweets (2022 test) with sentence-T5 plus GPT-Neo reranking. Hardalov et al. (2022) mine 333k noisy tweet–Snopes pairs from crowd replies and train sentence-BERT; on CheckThat! 2021 English test they report MAP@5 0.903 for the full system (vs Aschern 0.882 as quoted in their table; the lab overview reports 0.883).

Multilingual PFCR corpora and systems. Pikuliak et al. (2023) release MultiClaim (28k posts, 206k fact-checks, 31k pairs) and evaluate unsupervised and fine-tuned retrievers over all post–fact-check pairs with no held-out split, searching same-language fact-checks only; best monolingual S@10 is 0.83 (Generalizable T5-based dense Retriever, GTR-T5-Large, on English translations, averaged over languages with more than 100 pairs). Their post-newer vs post-older subsets score 0.800 vs 0.800 S@10 under the same index (no per-query date filter). Peng et al. (2025) organise SemEval-2025 Task 7 on MultiClaim with out-of-time test posts (pool to May 2024); winner PINGAN AI reaches monolingual average S@10 0.9601 and Polish 0.9260 on the hidden test languages. Devadiga et al. (2025) (a SemEval system) translate posts, fine-tune Stella with hard negatives and LLM-rerank; monolingual test S@10 0.938 (Polish 0.886). Ramponi et al. (2025) compare multilingual vs crosslingual retrieval on a curated MultiClaim v2 subset (47 languages) with an 80/10/10 language-stratified split that separates fact-checks across folds; LLM reranking of multilingual-e5-large reaches MRR@10 0.8042 multilingual. Vykopal et al. (2025) release AMC-16K (16k human Yes/No pairs from MultiClaim; 16% relevant overall) and evaluate LLMs as relevance detectors; best Macro-F1 0.825 on the monolingual half (Mistral Large few-shot).

Closest temporal precedents for PFCR. Shaar et al. (2022) add transcript context and a debate-level chronological train/test split on PolitiFact; the prior SOTA reranker drops from MAP 0.602 (sentence-level random) to 0.429 (debate chrono) without restricting the claim pool by date.

Time-respecting IR and verification (not PFCR index filtering). Ounis et al. (2011) define the 2011 Microblog track of the Text REtrieval Conference (TREC) real-time search with a compulsory run that bans future tweets and future corpus statistics; best compulsory P@30 is 0.4551. Liška et al. (2022) evaluate QA over a growing dated news corpus by quarter; dated dense passage retrieval (DPR) raises recall@20 on recent generated questions from 0.43 to 0.57. Glockner et al. (2022) show that a share of 0.697 of MultiFC misinformation claims have leaked fact-check evidence in search snippets (verification leakage from later fact-checks). Schlichtkrull et al. (2023) release AVeriTeC with evidence restricted to documents before the claim and temporally ordered splits; baseline verdict macro-F1 is 0.23 (gold evidence 0.49) on the development set; the test labels were not public. Mu et al. (2023) show rumour classifiers collapse under chronological splits: on Twitter 16, BERT macro-F1 falls from 0.915 (random) to 0.472 (stratified chronological).

Table metrics. Let $G(q)$ be the gold items for query q and r_q the rank of the first gold hit (undefined if none). *S@10* is the fraction of queries with a gold hit in the top 10; *MRR* is the mean of $1/r_q$ (0 if no gold ranked) (Pikuliak et al., 2023; Peng et al., 2025). *MRR@10* clips reciprocal rank at 10. *MAP@k* averages, over queries, precision at each relevant hit within the top k (plain MAP uses the full list). *P@k / R@k*: fraction of the top k that are relevant / fraction of all gold recovered in the top k. *Macro-F1*: unweighted mean of per-class F1 (classification, not ranking). *Leak rate*: share of claims whose later fact-check evidence already appears in search snippets (Glockner et al., 2022).

3 Datasets and status

Table 2 lists the source datasets and the two datasets we will create. For ranking corpora, class balance is the density of positive post–fact-check

Table 1: Published numbers for the fifteen works above. All values on a 0–1 scale; each names the dataset and the split or evaluation condition of the source paper. Metrics as above.

Work	Metric	Value	Dataset / split or condition
(Shaar et al., 2020)	MRR	0.788	Snopes tweets / test; BM25 top-50 + rankSVM
(Shaar et al., 2020)	S@10	0.915	same
(Shaar et al., 2021)	MAP@5	0.883	CheckThat! 2021 English Task 2A / test of the 70-15-15 split (Aschern; 0.882 in (Hardalov et al., 2022))
(Shaar et al., 2021)	MAP@5	0.749	same / BM25 baseline
(Nakov et al., 2022)	MAP@5	0.956	CheckThat! 2022 English Task 2A / 2022 test (RIET Lab)
(Hardalov et al., 2022)	MAP@5	0.903	CheckThat! 2021 English / test; CrowdChecked-trained system
(Pikuliak et al., 2023)	S@10	0.83	MultiClaim / all pairs, no split; same-language pool; GTR-T5-L on English text
(Pikuliak et al., 2023)	S@10	0.800 / 0.800	MultiClaim / post-newer vs post-older, same index, GTR-T5
(Peng et al., 2025)	S@10	0.9601	SemEval-2025 Task 7 / monolingual test average (PINGAN AI)
(Peng et al., 2025)	S@10	0.926	same / Polish test (PINGAN AI)
(Devadiga et al., 2025)	S@10	0.938	SemEval-2025 Task 7 / monolingual test (TIFIN)
(Devadiga et al., 2025)	S@10	0.886	same / Polish test (TIFIN)
(Ramponi et al., 2025)	MRR@10	0.804	MultiClaim v2 curated / test partition of the 80-10-10 split; LLM rerank
(Vykopal et al., 2025)	Macro-F1	0.825	AMC-16K / monolingual half, no split; Mistral Large few-shot
(Shaar et al., 2022)	MAP	0.602	PolitiFact debates / sentence-level random split
(Shaar et al., 2022)	MAP	0.429	same / debate-level chronological split
(Ounis et al., 2011)	P@30	0.455	TREC 2011 Microblog / 49 judged topics; compulsory no-future run
(Liška et al., 2022)	R@20	0.57 / 0.43	StreamingQA / 2020 recent generated questions; dated vs undated DPR
(Glockner et al., 2022)	leak rate	0.697	MultiFC / all misinformation claims, no split; automatic leak detector
(Schlichtkrull et al., 2023)	macro-F1	0.23 / 0.49	AVeriTeC / development set; baseline vs gold evidence
(Mu et al., 2023)	macro-F1	0.915 / 0.472	Twitter 16 / random vs stratified chronological split, BERT

links (gold pairs per post, share of fact-checks with a linked post), not a balanced classification split. Counts marked *to compute* come from the data itself in week 1; counts from the papers are marked with the source.

4 Concept, methodology and work plan

Pipeline. Figure 1 summarises the evaluation path. Systems (committed core, four): BM25; multilingual-e5-large (Wang et al., 2024); the Polish–English retrieval encoder mmlw-retrieval-roberta-large (chosen over LaBSE because it was trained on Polish–English pairs); bge-reranker-v2-m3 on the top 100 (Chen et al., 2024). Protocols: full-pool random; strict causal $P(q)$; size-matched random of $|P(q)|$. Compute: local open-weight models only; no commercial LLM API on post text.

Primary metric. The central claim is the Polish–English difference-in-differences of unconditional S@10 between the random and the strict causal pool (unanswerable queries count as misses). MRR and the answerable-only versions are secondary. The per-language answerable rate is the first result reported. All protocols are scored on one fixed query set per language, so the random, size-matched and causal pools differ only in the index.

Dates and ties. A fact-check or post may carry several dated instances, so we take the earliest instance at day resolution as its time; $P(q)$ uses the strict inequality $t_f < t_q$, so same-day pairs are excluded. Sensitivity runs use margins of 0, 1 and 7 days and a conservative and a liberal bound (latest vs earliest instance); we report the answerable rate and headline scores under both, and the share of queries whose pool membership flips. 100 dates per language are checked against live URLs; undated posts are dropped and counted. Fact-checks whose earlier sibling debunks are missing from the corpus are counted as answerable misses in the decomposition.

Table 2: Source datasets. FC = fact-checks: short verified/debunked claims from fact-checking sites (the documents we retrieve).

Dataset	Size	Label definition	Class balance	Licence	How obtained	Status
MultiClaim v1	206k FC, 28k posts, 31k pairs	fact-checker gold links (ClaimReview / warnings)	to compute (W1)	Zenodo access conditions, restricted record	Zenodo 7737983	READY
MultiClaim v2	435k FC, 89k posts, 105k pairs	same + relationship types	to compute (W1)	same	Zenodo 15413169	READY
SemEval-2025 T7	train from MultiClaim; test 4.3k mono posts (PL: 500 / 564 pairs)	shared-task gold for S@10	to compute (W1)	same	Zenodo 14989176	READY
MultiClaimNet	85.3k FC in 30.9k clusters	same-claim cluster ids	n/a (clustering)	CC BY 4.0	Zenodo 15100352	READY
AMC-16K	16k pairs / 20 langs	human Yes/No relevance	16% relevant overall; PL mono 20.25%	CC BY-NC-SA 4.0	authors (not via the Language Data Space, LDS)	RESTRICTED; request not yet sent, date logged on sending
Polish / English (papers: MultiClaim v1 posts, fact-checks; v2 subset monolingual pairs; SemEval test pool)						
PL	453 posts, 6,912 FC	as above	to compute (W1)	as above	as above	v2: 1,309 pairs; test pool 8,796
EN	7,307 posts, 85,814 FC	as above	to compute (W1)	as above	as above	v2: 19,341 pairs; test pool 145,287
Datasets we will create						
Causal split	id lists per language and protocol, no post text	post–fact-check ids with the date rule of Sec. 4	n/a	our release terms; id-list redistribution to confirm with the dataset authors	built by us, W1–W2	TO COLLECT
Failure annotations	50 queries (25 PL, 25 EN)	failure code per newly failing query (codebook by W3)	n/a	our own	Florek, Cwalina, by 16 Dec	TO COLLECT

Metrics. Chosen before results (MultiClaim / SemEval-2025 convention (Pikuliak et al., 2023; Peng et al., 2025)). Let $G(q)$ be the gold fact-checks for query q and r_q the rank of the first gold hit ($1/r_q = 0$ if none). *S@10* is the fraction of queries with $r_q \leq 10$; *MRR* is the mean of $1/r_q$. *Answerable rate* is the share of queries with $G(q) \cap P(q) \neq \emptyset$ under the causal pool. We report both unconditionally (unanswerable queries count as misses) and on answerable queries only; significance via paired cluster bootstrap; Polish–English contrast via difference-in-differences.

Concrete success deliverables. (1) Rebuildable causal split (id lists + code; no post text). (2) Shared results table: $\geq$4 systems $\times$ random vs causal $\times$ Polish and English, with bootstrap intervals; central number = extra Polish temporal cost. (3) Gap decomposition plus coded analysis of 50 newly failing queries.

Week-by-week plan. Internal work finishes before Christmas; course checkpoints: mid-term video 18 Nov, Phase 2 2 Dec, formal final 13 Jan 2027. One owner per task.

- **W1 (to 18 Nov):** download MultiClaim v1/v2 and SemEval Task 7, dated corpus loader (Pozorski); causal-pool module and answerable-rate gate (Florek); AMC-16K access request (Pozorski).
- **W2 (to 25 Nov):** BM25 and multilingual-e5 on the random pool, PL and EN (Jarosz); mmlw and reranker, GPU runs (Florek); per-language answerable rates (Pozorski).
- **W3 (to 2 Dec):** causal and size-matched runs for all four systems (Jarosz); Phase 2 report and proof-of-concept (PoC) table (Cwalina).
- **W4 (to 9 Dec):** bootstrap intervals and difference-in-differences (Florek); gap decomposition with covariates (Pozorski).
- **W5 (to 16 Dec):** 50 failure annotations and codebook (Cwalina); judge validation (Jarosz).
- **W6 (to 23 Dec):** release check of split and code (Pozorski); write-up (Cwalina).

Main risks: unknown Polish answerable rate (if $<$150, Polish–English contrast becomes descriptive); timestamp semantics undocumented (URL

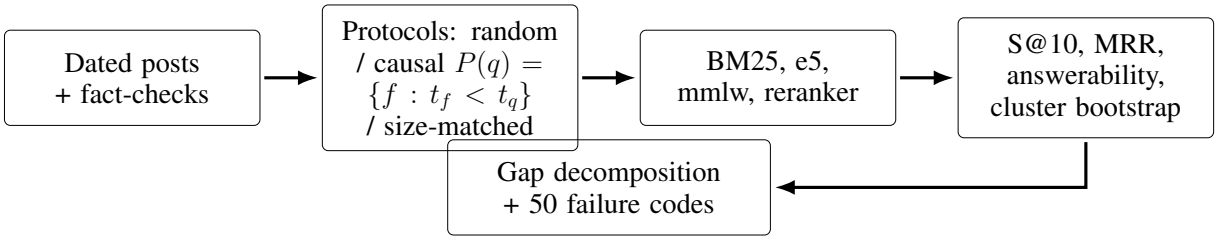


Figure 1: Evaluation pipeline.

date audit + margin sensitivity); AMC-16K still to be requested.

5 Data statement

Primary data: MultiClaim v1/v2 and SemEval-2025 Task 7 from restricted Zenodo records (research use only; no re-identification or re-sharing beyond named requesters) (Moro et al., 2025; Pikuliak et al., 2023; Peng et al., 2025); MultiClaimNet (Zenodo; CC BY 4.0) (Panchendrarajan et al., 2025). Access is granted for MultiClaim v1/v2, SemEval-2025 Task 7 and MultiClaimNet. Posts are real social-media text, released in anonymised form by the dataset authors; we treat them as personal data. We store files only in a directory readable by named team members, never send post text to an external LLM API, never put raw text in git or the paper (examples paraphrased; errors aggregated), and delete the data at project end. No per-outlet or per-author error ranking will be published. AMC-16K is secondary: it validates the LLM relevance judges used to label sibling fact-checks and failure codes. The judges are a panel of three open-weight models disjoint from the evaluated systems (gpt-oss-120b, Command A, Mistral Large 2411), prompted in English with the Yes/No labels of AMC-16K and validated on its 400 Polish pairs; agreement with the human labels is Cohen's κ, and a judge is used only above $\kappa = 0.6$ (threshold to confirm with the supervisor). Its public listing is on the EU Language Data Space (LDS) catalogue, which our university cannot use; we will request a copy from the authors at the Kempelen Institute of Intelligent Technologies (KInIT) under research terms.

6 Contribution

Table 3 lists each member's tasks and approximate hours spent preparing this report.

Table 3: Contributions to preparing this report. Hours are approximate.

Member	Role focus	Hours
Paweł Pozorski	data, protocol & SOTA	7
Mateusz Jarosz	systems & SOTA	7
Paweł Florek	evaluation, figure & LaTeX	5
Jan Cwalina	drafting & polish	5

7 LLM-use declaration

We used the following AI tools while preparing this proposal and report: Claude Code (Anthropic Claude models) [TODO: every member lists each tool and model version used, e.g. ChatGPT (GPT-x), Overleaf AI]. Their role was limited to formatting (LaTeX tables, section structure, citation layout) and wording corrections (clarity, grammar, consistency of terminology).

References

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335.

Prasanna Devadiga, Arya Suneesh, Pawan Rajpoot, Bharatdeep Hazarika, and Aditya Baliga. 2025. TIFIN india at SemEval-2025: Harnessing translation to overcome multilingual IR challenges in fact-checked claim retrieval. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*, pages 2297–2304.

Max Glockner, Yufang Hou, and Iryna Gurevych. 2022. Missing counter-evidence renders NLP fact-checking unrealistic for misinformation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 5916–5936.

Momchil Hardalov, Anton Chernyavskiy, Ivan Koychev, Dmitry Ilvovsky, and Preslav Nakov. 2022.

CrowdChecked: Detecting previously fact-checked claims in social media. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 266–285.

Adam Liška, Tomáš Kočiský, Elena Gribovskaya, Tayfun Terzi, Eren Sezener, Devang Agrawal, Cyprien de Masson d'Autume, Tim Scholtes, Manzil Zaheer, Susannah Young, Ellen Gilsenan-McMahon, Sophia Austin, Phil Blunsom, and Angeliki Lazaridou. 2022. StreamingQA: A benchmark for adaptation to new knowledge over time in question answering models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 13604–13622.

Robert Moro, Ivan Srba, Matúš Pikuliak, Martin Melišek, Juraj Podroužek, Matúš Mesarčík, Natália Slosiarová, Jakub Simko, and Maria Bielikova. 2025. MultiClaim: Multilingual previously fact-checked claim retrieval (dataset records v1 and v2). Zenodo. Restricted-access dataset records.

Yida Mu, Kalina Bontcheva, and Nikolaos Aletras. 2023. It's about time: Rethinking evaluation on rumor detection benchmarks using chronological splits. In *Findings of the Association for Computational Linguistics: EACL 2023*, pages 736–743.

Preslav Nakov, Giovanni Da San Martino, Firoj Alam, Shaden Shaar, Hamdy Mubarak, and Nikolay Babulkov. 2022. Overview of the CLEF-2022 CheckThat! lab task 2 on detecting previously fact-checked claims. In *Working Notes of CLEF 2022*, pages 393–403. CEUR Workshop Proceedings Vol-3180.

Iadh Ounis, Craig Macdonald, Jimmy Lin, and Ian Soboroff. 2011. Overview of the TREC-2011 microblog track. In *Proceedings of the Twentieth Text REtrieval Conference (TREC 2011)*. NIST Special Publication 500-296.

Rrubaa Panchendrarajan, Rubén Míguez Pérez, and Arkaitz Zubiaga. 2025. MultiClaimNet: A massively multilingual dataset of fact-checked claim clusters. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 11203–11215.

Qiwei Peng, Robert Moro, Michal Gregor, Ivan Srba, Simon Ostermann, Marian Simko, Juraj Podroužek, Matúš Mesarčík, Jaroslav Kopčan, and Anders Søgaard. 2025. SemEval-2025 task 7: Multilingual and crosslingual fact-checked claim retrieval. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*, pages 2498–2511.

Matúš Pikuliak, Ivan Srba, Robert Moro, Timo Hromadka, Timotej Smoleň, Martin Melišek, Ivan Vykopal, Jakub Simko, Juraj Podroužek, and Maria Bielikova. 2023. Multilingual previously fact-checked claim retrieval. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 16477–16500.

Alan Ramponi, Marco Rovera, Robert Moro, and Sara Tonelli. 2025. Multilingual vs crosslingual retrieval of fact-checked claims: A tale of two approaches. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 29045–29064.

Michael Schlichtkrull, Zhijiang Guo, and Andreas Vlachos. 2023. AVeriTeC: A dataset for real-world claim verification with evidence from the web. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*.

Shaden Shaar, Nikolay Babulkov, Giovanni Da San Martino, and Preslav Nakov. 2020. That is a known lie: Detecting previously fact-checked claims. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3607–3618.

Shaden Shaar, Fatima Haouari, Watheq Mansour, Maram Hasanain, Nikolay Babulkov, Firoj Alam, Giovanni Da San Martino, Tamer Elsayed, and Preslav Nakov. 2021. Overview of the CLEF-2021 CheckThat! lab task 2 on detecting previously fact-checked claims in tweets and political debates. In *Working Notes of CLEF 2021*, pages 393–405. CEUR Workshop Proceedings Vol-2936.

Shaden Shaar, Firoj Alam, Giovanni Da San Martino, and Preslav Nakov. 2022. The role of context in detecting previously fact-checked claims. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 1619–1631.

Anders Søgaard, Sebastian Ebert, Jasmijn Bastings, and Katja Filippova. 2021. We need to talk about random splits. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 1823–1832.

Ivan Vykopal, Matúš Pikuliak, Simon Ostermann, Tatiana Anikina, Michal Gregor, and Marian Simko. 2025. Large language models for multilingual previously fact-checked claim detection. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 15741–15765.

Ivan Vykopal, Antonia Karamolegkou, Jaroslav Kopčan, Qiwei Peng, Tomáš Javůrek, Michal Gregor, and Marian Simko. 2026. Investigating language and retrieval bias in multilingual previously fact-checked claim detection. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5195–5221.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 text embeddings: A technical report. Technical Report 2402.05672, arXiv. Preprint technical report.